

CORN3000-EI

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Invalid Date

Table of contents

Preface

Here you will find course materials for the module CORN3000 Environmental Investigations. This is part of the BSc Environmental Resource Management, delivered at CCUC: Newquay. This qualification is validated by Plymouth University.

1 Gannel ERM 2025-2026 analysis

Analysis of data from file trip to the Gannel 2/3/2026 with ERM group.

1.1 Load packages

```
library(tidyverse)
library(here)
library(janitor) # cleans names of columns
library(readxl)
library(cowplot)
```

1.2 Load data

```
filepath <- here("data","gannel_2526_data.xlsx")
gannel <- read_excel(filepath,sheet = "Sediment")
# glimpse(gannel)
```

1.3 Organic matter per cubic metre of wet sediment

We want to calculate the mass of organic matter per cubic metre of sediment. This is given by

$$\text{kg OM/m}^3 = \rho_{ws} \times p_{ds} \times p_{om}$$

where ρ_{ws} is the density of the wet sediment, p_{ds} is the proportion of dry sediment in the wet sediment and p_{om} is the proportion of organic matter in the dry sediment.

We will calculate each of these terms separately